

Disproof of Conjecture 00000001685: no even subdivision of $K_{3,3}$ is a Pfaffian graph

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Abstract

Conjecture 00000001685 says that the minimal families of nonplanar Pfaffian graphs are the Möbius ladders M_{2k} and the even subdivisions of $K_{3,3}$. In particular it says that the even subdivisions of $K_{3,3}$ are Pfaffian graphs. This is false. We prove that *no* even subdivision of $K_{3,3}$ has a Pfaffian orientation, including $K_{3,3}$ itself, which is also the Möbius ladder on 6 vertices. The proof is a parity argument: in every orientation an odd number of the six perfect-matching terms of the Pfaffian are negative, so the terms never all have the same sign. The obstruction, and its instances for $K_{3,3}$, for a proper even subdivision of $K_{3,3}$, and for the 6-vertex Möbius ladder, are formalized in Lean 4 (core library only). As a secondary result we give a nonplanar Pfaffian graph of crossing number 1 that lies in neither family, and we show that the Möbius clause also fails if the families are read as non-Pfaffian obstructions.

1 The conjecture and the clause we refute

Definition: A Pfaffian orientable graph is one whose perfect matchings can be counted by signed cancellation. *Conjecture:* The minimal families of nonplanar Pfaffian graphs are the Möbius ladders M_{2k} and the even subdivisions of $K_{3,3}$; these two families exhaust the nonplanar Pfaffian class of crossing number ≤ 2 .

The Chinese version says the same. The first sentence names two families *of nonplanar Pfaffian graphs*, and the second calls them part of “the nonplanar Pfaffian class”. So the conjecture asserts, at the least:

Clause (E). Every even subdivision of $K_{3,3}$ is a (nonplanar) Pfaffian graph.

The conjecture is a conjunction that includes (E), so refuting (E) refutes the conjecture. Whatever “minimal” means, a family of nonplanar Pfaffian graphs has to consist of Pfaffian graphs, and the falsity of (E) does not depend on that word.

Readings covered.

- *Even subdivision.* An even subdivision of H replaces every edge by a path with an even number $(0, 2, 4, \dots)$ of internal vertices. We prove (Theorem 7) that *every* even subdivision of $K_{3,3}$ is non-Pfaffian. So it makes no difference whether $K_{3,3}$ itself counts as an even subdivision, whether at least one edge must be subdivided, or whether every edge must be subdivided.

- *Pfaffian*. We use the conjecture’s own definition (the signed sum counts the perfect matchings, $|\text{Pf}(A_D)| = \#\text{PM}$) and Kasteleyn’s standard definition (all perfect matchings have the same sign). They are equivalent (Lemma 2), and both are refuted.
- M_{2k} . If M_{2k} is the Möbius ladder on $2k$ vertices (k rungs), then $M_6 \cong K_{3,3}$ is not Pfaffian, so the Möbius part fails as well. More generally, the computations below show that the ladders with 3, 5 and 7 rungs are non-Pfaffian. If instead M_{2k} has $2k$ rungs, the ladders with 2, 4 and 6 rungs are Pfaffian, and we make no claim against the Möbius part. In both cases the refutation of (E) does not depend on this choice.
- *Obstruction reading*. Some readers may take the “minimal families” to be minimal *non*-Pfaffian obstructions, in the spirit of Little’s theorem [1]. Under that reading the Möbius clause fails in both conventions. The Wagner graph V_8 is Pfaffian (Lean: `M8_pfaffian`), and it lies in the family M_{2k} either way: $k = 4$ when counting vertices, and $k = 2$ when counting rungs. Lean: `obstruction_reading_false` (rung count, $M_{2k} = \text{mobius}(2k)$) and `obstruction_reading_false_vertex_convention` ($M_{2k} = \text{mobius } k$).

2 Definitions

Let G be a loopless graph on the vertex set $\{1, \dots, n\}$ (we use $\{0, \dots, n-1\}$ in the code). A *perfect matching* is a set of edges covering every vertex exactly once. An *orientation* D directs every edge, and its skew adjacency matrix is $A_D = (a_{ij})$, where $a_{ij} = 1$ if $i \rightarrow j$, $a_{ij} = -1$ if $j \rightarrow i$, and $a_{ij} = 0$ otherwise. Let $M = \{t_1 h_1, \dots, t_m h_m\}$ be a perfect matching ($n = 2m$), with each edge written tail-first ($t_k \rightarrow h_k$). Its *term* is

$$\text{sgn}_D(M) = \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2m-1 & 2m \\ t_1 & h_1 & \cdots & t_m & h_m \end{pmatrix} = (-1)^{\text{inv}(t_1 h_1 \cdots t_m h_m)},$$

where inv counts inversions. This does not depend on the order in which the edges are listed: moving a block of two entries past another entry is an even permutation. The Pfaffian is

$$\text{Pf}(A_D) = \sum_{\pi} \text{sgn}(\pi) \prod_k a_{\pi(2k-1)\pi(2k)} = \sum_M \text{sgn}_D(M).$$

Here π runs over the permutations with $\pi(1) < \pi(3) < \dots$ and $\pi(2k-1) < \pi(2k)$. A term is non-zero exactly when the pairs form a perfect matching M . Writing each pair tail-first changes the sign of the permutation and of the factor a together, so the term equals $\text{sgn}_D(M)$.

Definition 1. D is a *Pfaffian orientation* if $|\text{Pf}(A_D)|$ equals the number $\Phi(G)$ of perfect matchings. G is *Pfaffian* if it has a Pfaffian orientation.

Lemma 2. $|\text{Pf}(A_D)| = \Phi(G)$ if and only if all perfect matchings have the same sign $\text{sgn}_D(M)$.

Proof. Suppose p of the Φ terms are $+1$ and q are -1 . Then $|\text{Pf}| = |p - q|$, and this equals $p + q$ if and only if $pq = 0$. $\square$

3 The parity obstruction

Lemma 3. Reversing one edge e of D changes $\text{sgn}_D(M)$ for exactly the perfect matchings $M \ni e$.

Proof. If $e \notin M$, the word $t_1 h_1 \cdots t_m h_m$ is unchanged. If $e = t_k h_k \in M$, the two adjacent entries t_k, h_k are swapped. They are distinct because G is loopless, so the swap is a transposition and the inversion number changes by exactly 1. $\square$

Theorem 4 (Parity obstruction). *Let G be loopless, and suppose that*

- (a) $\Phi(G)$ is even;
- (b) every edge lies in an even number of perfect matchings;
- (c) for some orientation D_0 , the number of perfect matchings M with $\text{sgn}_{D_0}(M) = -1$ is odd.

Then G is not Pfaffian.

Proof. Let $\nu(D)$ be the number of negative terms. Any orientation D is obtained from D_0 by reversing edges one at a time. By Lemma 3, reversing e changes ν by an amount congruent modulo 2 to the number of perfect matchings containing e , which is even by (b). Hence $\nu(D) \equiv \nu(D_0) \equiv 1 \pmod{2}$ for every D . If D were Pfaffian, Lemma 2 would give $\nu(D) \in \{0, \Phi\}$, which is even by (a). This is a contradiction. $\square$

Equivalently, by (b) the product $\prod_M \text{sgn}_D(M) = (-1)^{\nu(D)}$ does not depend on D . It is -1 , while a Pfaffian orientation would make it $(\pm 1)^\Phi = 1$. This is the classical reason why $K_{3,3}$ is not Pfaffian (Little [1]; see also [3, 4]).

4 $K_{3,3}$

Let the vertices be $0, 1, 2$ and $3, 4, 5$, and let D_0 orient every edge $a \rightarrow b$ with $a \in \{0, 1, 2\}$. The perfect matchings are $M_\sigma = \{0\sigma(0), 1\sigma(1), 2\sigma(2)\}$ for the six bijections $\sigma : \{0, 1, 2\} \rightarrow \{3, 4, 5\}$. In the word $0\sigma(0)1\sigma(1)2\sigma(2)$ the entries $\sigma(i) \geq 3$ form $2 + 1 = 3$ inversions with the later entries $1, 2$, plus the inversions of σ itself, so $\text{sgn}_{D_0}(M_\sigma) = -\text{sgn}(\sigma)$:

matching	mask (Lean)	$\text{sgn}(\sigma)$	sgn_{D_0}
$\{05, 14, 23\}$	84	-1	$+1$
$\{04, 15, 23\}$	98	$+1$	-1
$\{05, 13, 24\}$	140	$+1$	-1
$\{03, 15, 24\}$	161	-1	$+1$
$\{04, 13, 25\}$	266	-1	$+1$
$\{03, 14, 25\}$	273	$+1$	-1

(Bit i of a mask selects edge number i in the order $03, 04, 05, 13, 14, 15, 23, 24, 25$.) Thus $\Phi = 6$; each edge ab lies in exactly 2 matchings (the 2 bijections with $\sigma(a) = b$); and $\nu(D_0) = 3$ is odd. By Theorem 4:

Proposition 5. $K_{3,3}$ is not Pfaffian. Concretely, $\text{Pf}(A_{D_0}) = 0$, and every one of the $2^9 = 512$ orientations has $|\text{Pf}(A_D)| \leq 4 < 6$.

(Indeed $\nu(D)$ is odd, so $|\text{Pf}| = |6 - 2\nu| \in \{4, 0\}$.)

5 All even subdivisions of $K_{3,3}$

Let G have vertices $\{0, \dots, n-1\}$ and an edge uv . Let G' be the graph obtained by replacing uv with the path $uabv$, where $a = n$ and $b = n+1$ are new vertices. Every even subdivision of $K_{3,3}$ arises from $K_{3,3}$ by finitely many such steps, up to relabelling of the vertices.

Lemma 6. *If G satisfies (a), (b) and (c) of Theorem 4, so does G' , and $\Phi(G') = \Phi(G)$.*

Proof. In G' the vertex a is matched either to b or to u ; in the latter case b is matched to v . This gives a bijection $M \mapsto M'$ from the perfect matchings of G :

$$M' = M \cup \{ab\} \text{ if } uv \notin M, \quad M' = (M \setminus \{uv\}) \cup \{ua, bv\} \text{ if } uv \in M.$$

So $\Phi(G') = \Phi(G)$ and (a) is inherited. Let c be the number of perfect matchings of G through uv . In G' the edges ua and bv lie in c perfect matchings and ab lies in $\Phi - c$. Both numbers are even, and the other edges keep their counts, so (b) is inherited. For (c): since G satisfies (b), the proof of Theorem 4 shows that $\nu(D)$ is odd for *every* orientation D of G , so we may take D_0 with $u \rightarrow v$. Orient $u \rightarrow a \rightarrow b \rightarrow v$ and keep every other edge. If $uv \notin M$, the word of M' is the word of M followed by ab . These are the two largest labels, in increasing order, so the sign is unchanged. If $uv \in M$, list uv last, so that the word of M is Wuv . The word of M' is $Wuabv$. It arises from $Wuvab$ (which has the sign of Wuv) by moving v past a and b , an even permutation. Hence $\text{sgn}_{D'_0}(M') = \text{sgn}_{D_0}(M)$ for all M , and ν is preserved. $\square$

Relabelling the vertices by a permutation τ multiplies every term by $\text{sgn}(\tau)$. So conditions (a), (b) and (c) are invariant under relabelling: when Φ is even, ν either stays the same or becomes $\Phi - \nu$. Being Pfaffian is also invariant. Induction with Lemma 6, starting from Proposition 5, gives:

Theorem 7. *Every even subdivision of $K_{3,3}$ (including $K_{3,3}$ itself) satisfies the hypotheses of Theorem 4 with $\Phi = 6$. Hence no even subdivision of $K_{3,3}$ is Pfaffian, and Clause (E) of Conjecture 00000001685 is false.*

Corollary 8. *Conjecture 00000001685 is false.*

The smallest proper even subdivision is S : $K_{3,3}$ with edge 03 replaced by the path 0673 (8 vertices, 11 edges). It is non-Pfaffian with $\Phi(S) = 6$. The masks of its perfect matchings, over the edge order 04, 05, 13, 14, 15, 23, 24, 25, 06, 67, 73, are 554, 561, 582, 645, 1360, 1416.

6 Möbius ladders

The Möbius ladder with r rungs has vertices $0, \dots, 2r-1$, the cycle edges $i(i+1 \bmod 2r)$ and the rungs $i(i+r)$. For $r = 3$ it is $K_{3,3}$, with colour classes the even and the odd vertices. Its 6 perfect matchings satisfy the obstruction directly; this is proved in Lean as `M6_not_pfaffian`. An exhaustive computation in `verify.py`, using the method of the next section, gives the following:

rungs r	2	3	4	5	6	7
vertices	4	6	8	10	12	14
Pfaffian?	yes	no	yes	no	yes	no

The ladder with 5 rungs has 13 perfect matchings, and none of its 2^{15} orientations is Pfaffian. In the vertex-count convention, then, the Möbius part of the conjecture also fails (for example at M_6 and M_{10}). In the rung-count convention we make no claim about it.

7 Secondary: the second sentence

Read the second sentence as “every nonplanar Pfaffian graph of crossing number ≤ 2 is a Möbius ladder or an even subdivision of $K_{3,3}$ ”. Under this reading it is also false. Let W be the Wagner graph V_8 (the Möbius ladder with 4 rungs) with the edge 01 replaced by a path 0891. Then:

- W is Pfaffian, with $\Phi(W) = 7$. The proof of Lemma 6 shows that the compatible orientation of a Pfaffian orientation of V_8 is Pfaffian. The Lean theorem `M8.pfaffian` exhibits a Pfaffian orientation of V_8 , and `verify.py` checks W directly.
- W is nonplanar. $V_8 - \{04\}$ is a subdivision of $K_{3,3}$ with classes $\{1, 3, 6\}$ and $\{2, 5, 7\}$: the paths are 12, 15, 107, 32, 37, 345, 65, 67, 62.
- $\text{cr}(W) = 1$. The Wagner graph has crossing number $\text{cr}(V_8) = 1$: Guy and Harary [2] show that every Möbius ladder with at least 3 rungs has crossing number 1. Crossing number is invariant under subdividing edges: a drawing of G is a drawing of any subdivision of G with the same crossings, and conversely, by suppressing the degree-2 vertices.
- W is not a Möbius ladder, since it has vertices of degree 2. It is not an even subdivision of $K_{3,3}$, since $|E| - |V| = 14 - 10 = 4$, whereas every subdivision of $K_{3,3}$ has $|E| - |V| = 3$.

This part is not formalized in Lean, apart from `M8.pfaffian`. The main disproof does not depend on it.

8 Independent computational check (verify.py)

The script uses only the Python standard library and a different method from the Lean proof. Since $\det A_D = \text{Pf}(A_D)^2$ and $|\text{Pf}| \leq \Phi$, G is Pfaffian if and only if some orientation has $\det A_D = \Phi^2$. The script computes exact integer determinants (Bareiss elimination) and enumerates perfect matchings recursively. It checks:

- $K_{3,3}$ (all 512 orientations), S (all 2048), and $K_{3,3}$ with two edges subdivided twice (all 8192): none is Pfaffian, and $\Phi = 6$ in each case.
- $K_{3,3}$ with every edge subdivided 2 times, and with every edge subdivided 4 times. Switching at a vertex (reversing all its edges) multiplies the Pfaffian by -1 and leaves $\det$ unchanged. So it suffices to check one orientation per switching class: fix the edges of a spanning tree, which leaves $2^{|E|-|V|+1} = 16$ orientations. None is Pfaffian.
- The hypotheses (a), (b), (c) of Theorem 4 for all 2^9 even subdivisions with 0 or 2 internal vertices per edge, and for three more with up to 6.
- Möbius ladders with 2–7 rungs (the table above), with 5 rungs also exhaustively over 2^{15} orientations; and the graph W of the previous section.

9 Lean formalization

`lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). Everything is defined from scratch:

- **Graph** is a vertex count and an edge list (edge endpoints are assumed to be $< n$, which holds for every graph used). **sel es M** is the edge subset selected by the bit mask M . **isPM** says every vertex $v < n$ lies on exactly one selected edge. **PMs G** lists all perfect matchings among the $2^{|E|}$ masks, and **numPM G** is their number.
- An orientation is a **List Bool**, one entry per edge (**true** keeps (a, b) as $a \rightarrow b$). **par** is the inversion parity of the word $t_1 h_1 t_2 h_2 \dots$, and **term** is ± 1 accordingly. **Pf G os** = $\sum_M \text{term}$.
- **IsPfaffian G**: some orientation of length $|E|$ has $|\text{Pf}| = \text{numPM}$ (the conjecture’s definition). **IsPfaffian’**: all terms are equal (Kasteleyn’s definition). **same_sign_pfaffian** shows that the second implies the first.
- **not_pfaffian_of_invariant** is Theorem 4 for arbitrary graphs. It is proved through **word_perm**, **par_flips** (Lemma 3 in cumulative form: the parity changes by the number of reversed matching edges), **sum_map_flips_even** and **sum_terms**.
- Computations (kernel **decide**): **PMs_K33**, **PMs_S**, **PMs_M6**, and the hypotheses (a)–(c). The results are **K33_not_pfaffian**, **S_not_pfaffian** and **M6_not_pfaffian**.
- **EvenSubdivisionK33** is the inductive family generated from $K_{3,3}$ by **subdivide2**, which replaces an edge uv by $un(n+1)v$. **S_even_subdivision** shows S belongs to it and $S \neq K_{3,3}$.
- Main theorems: **conjecture_00000001685_false (Q)**: $\neg \forall G, \text{EvenSubdivisionK33 } G \rightarrow Q \wedge \text{IsPfaffian } G$, for an arbitrary extra property Q such as “non-planar” or “minimal”; the same with **IsPfaffian’**; **false_proper** (members other than $K_{3,3}$); **_conjunction_false** ($\neg((E) \wedge R)$ for any R); and **mobius_clause_false_vertex_convention**; and for the obstruction reading, **obstruction_reading_false** and **obstruction_reading_false_vertex_convention**.
- Non-vacuity: **K4_pfaffian** (K_4 , the ladder with 2 rungs) and **M8_pfaffian** (the nonplanar Wagner graph) have explicit Pfaffian orientations, so **IsPfaffian** is not empty. As a cross-check that **Pf** is the matrix Pfaffian, **pfMat_K33** and **pfMat_M8** compare it with the recursive row expansion $\text{Pf}(A) = \sum_j (-1)^j a_{1j} \text{Pf}(A_{\hat{1}\hat{j}})$ of the skew matrix.

Not in Lean. Lean covers $K_{3,3}$ and S . These suffice for every reading in which the clause applies to all even subdivisions (with or without $K_{3,3}$ itself). The induction to all even subdivisions (Lemma 6) is proved on paper only, as is the invariance under relabelling, and Section 7 apart from **M8_pfaffian**. There is no **sorry**, no **native_decide**, and no added axiom. **#print axioms** shows at most **propext** and **Quot.sound**.

10 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only (about 1 minute)
cd .. && python3 verify.py # Python 3 standard library
pdflatex report.tex && pdflatex report.tex
```

References

- [1] C. H. C. Little, A characterization of convertible $(0, 1)$ -matrices, *J. Combin. Theory Ser. B* 18 (1975), 187–208.

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- [3] L. Lovász and M. D. Plummer, *Matching Theory*, North-Holland, 1986.
- [4] N. Robertson, P. D. Seymour and R. Thomas, Permanents, Pfaffian orientations, and even directed circuits, *Ann. of Math.* 150 (1999), 929–975.